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case 1:

$$W - F_E = ma$$

$$W - q \frac{V}{x} = ma$$

case 2:

$$W + F_E = ma$$

$$W + q \frac{V}{x} = m(5a)$$

$$(2) - (1): 2q \frac{V}{x} = 4ma$$

$$q = 2 \frac{max}{V}$$

**23**

$$I = nqvA$$

$$= (8.2 \times 10^{28})(1.6 \times 10^{-19})(2.3 \times 10^{-5})(8.0 \times 10^{-6})$$